

PUC

(Nested Loops – while and
for, Jumps in Loops)

WEEK – 05 – 01

(Practice Session – Coding)

Status : Finished

Started : Monday, 23 December 2024, 5:33 PM

Completed : Friday, 22 November 2024, 12:31 PM

Duration : 31 days 5 hours

Question 1

Correct

Marked out of 3.00

Flag question

Objective:

Write a program that prints a simple chessboard.

Input format:

The first line contains the number of inputs T.

The lines after that contain a different values for size of the chessboard

Output format:

Print a chessboard of dimensions size size. Print W for white spaces and B for blackspace

Input:

2 3 5

Output:

WBW

BWB

WBW

WBWBW

BWBWB

WBWBW

BWBWB

WBWBW

```
1  #include<stdio.h>
2  int main(){
3      int n,a;
4      scanf("%d",&n);
5      while(n!=0)
6  {
7      scanf("%d",&a);
8      for(int i=0;i<a;i++){
9          for(int j=0;j<a;j++){
10             if((i+j)%2==0)
11                 printf("W");
12             else
13                 printf("B");
14             }
15             printf("\n");
16         }
17         n--;
18     }}
```

	Input	Expected	Got	
✓	2	WBW	WBW	✓
	3	BWB	BWB	
	5	WBW	WBW	
		WBWBW	WBWBW	
		BWBWB	BWBWB	
		WBWBW	WBWBW	
		BWBWB	BWBWB	
		WBWBW	WBWBW	

Passed all tests! ✓

Question 2

Correct

Marked out of 5.00

Flag question

Objective :

Let's print a chessboard!

Write a program that takes input:

The first line contains T, the number of test cases

Each test case contains an integer N and also the starting character of the chessboard

Output Format

Print the chessboard as per the given examples

Sample Input/Output

Input:

2

2 W

3 B

Output:

WB

BW

BWB

WBW

BWB

Answer: (penalty regime: 0%)

```
1 #include<stdio.h>
2 int main(){
3     int n,a;
4     char ch;
5     scanf("%d",&n);
6     while(n!=0){
7         scanf("%d %c",&a,&ch);
8         for(int i=0;i<a;i++){
9             for(int j=0;j<a;j++){
10                 if(ch=='W'){
11                     if((i+j)%2==0)
12                         printf("W");
13                     else
14                         printf("B");}else{
15                     if((i+j)%2==0)
16                         printf("B");
17                     else
18                         printf("W");
19                 }
20             }
21             printf("\n");
22         }
23         n--;
24     }
25 }
```

	Input	Expected	Got	
✓	2	WB	WB	✓
	2 W	BW	BW	
	3 B	BWB	BWB	
		WBW	WBW	
		BWB	BWB	

Passed all tests! ✓

Question 3

Correct

Marked out of 7.00

Flag question

Objective :

Decode the logic and print the Pattern that corresponds to given input.

If N= 3

Then pattern will be:

10203010011012

**4050809

****607

If N= 4, then pattern will be:

1020304017018019020

**50607014015016

****809012013

Constraints.

$2 \leq N \leq 100$

Input Format

First line contains T, the number of test cases

Each test case contains a single integer N

Output

First line print Case #i where i is the test case number In the subsequent line, print the pattern.

Test Case 1

3

3

4

5

Output

Case #1

10203010011012

**4050809

****607

Case #2

1020304017018019020

**50607014015016

****809012013

Case #3

102030405026027028029030

**6070809022023024025

****10011012019020021

*****13014017018

*****15016

Answer: (penalty regime: 0%)

```

1  #include<stdio.h>
2  int main(){
3      int t,n,x,y,z=1,i,ans,c;
4      scanf("%d",&t);
5      while(z<=t){
6          scanf("%d",&n);
7          printf("Case #%d\n",z);
8          y=1;
9          i=1;
10         c=0;
11         while(y<=n){
12             x=1;
13             ans=(n*n);
14             ans=ans-c;
15             while(x<=2*n){
16                 if(x<=n){
17                     if(x<y)
18
19                 printf("**");
20             else if(x<=n){
21                 printf("%d",i*10);
22                 i++;
23             }}else{
24                 if((x+y)==(2*n)+1){
25                     printf("%d",(ans+y));
26                     ans++;
27                     c++;
28
29                 }
30             else if(x+y<=(2*n)+1){
31                 printf("%d",(ans+y)*10);
32                 ans++;
33                 c++;
34             }
35         }
36         x++;
37
38         }
39         y++;
40         printf("\n");
41     }
42     z++;
43 }

```

	Input	Expected	Got	
✓	3	Case #1	Case #1	✓
	3	10203010011012	10203010011012	
	4	**4050809	**4050809	
	5	****607	****607	
		Case #2	Case #2	
		1020304017018019020	1020304017018019020	
		**50607014015016	**50607014015016	
		****809012013	****809012013	
		*****10011	*****10011	
		Case #3	Case #3	
		102030405026027028029030	102030405026027028029030	
		**6070809022023024025	**6070809022023024025	
		****10011012019020021	****10011012019020021	
		*****13014017018	*****13014017018	
		*****15016	*****15016	

Passed all tests! ✓

WEEK – 05 – 02

(Practice Session – Coding)

Status : Finished

Started : Monday, 23 December 2024, 5:33 PM

Completed : Friday, 22 November 2024, 1:07 PM

Duration : 31 days 4 hours

Question 1

Correct

Marked out of 3.00

Flag question

Objective :

The k-digit number N is an Armstrong number if and only if the k-th power of each digit sums to N.

Given a positive integer N, return true if and only if it is an Armstrong number.

Example 1:

Input:

153

Output:

True

Explanation:

153 is a 3-digit number, and $1^3 + 5^3 + 3^3 = 153$.

Example 2:

Input:

123

Output:

False

Explanation:

123 is a 3-digit number, and $1^3 + 2^3 + 3^3 = 36$.

Example 3:

Input:

1634

Output:

True

Note:

$1 \leq N \leq 10^8$

Answer: (penalty regime: 0%)

```
1  #include<stdio.h>
2  #include<math.h>
3  int main(){
4      int n,a,p=0,t,sum=0,x;
5      scanf("%d",&n);
6      t=n;
7      a=n;
8      while(t!=0){
9          t=t/10;
10         p++;
11     }
12     while(n!=0)
13     {
14         x=n%10;
15         sum=sum+pow(x,p);
16         n=n/10;
17     }
18     if(sum==a)
19
20     printf("true");
21     else
22     printf("false");
23 }
```

	Input	Expected	Got	
✓	153	true	true	✓
✓	123	false	false	✓

Passed all tests! ✓

Question 2

Correct

Marked out of 5.00

Flag question

Objective :

Take a number, reverse it and add it to the original number until the obtained number is a palindrome. Constraints $1 \leq \text{num} \leq 99999999$

Sample Input

1 32

Sample Output

1 55

Sample Input

2 789

Sample Output

2 66066

Answer: (penalty regime: 0%)

```
1 #include<stdio.h>
2 int main(){
3     long long int n,s,r,t,ts;
4     scanf("%lld",&n);
5     while(1){
6         r=0;
7         t=n;
8         while(n){
9             r=r*10+(n%10);
10            n=n/10;
11
12        }
13        s=t+r;
14        ts=s;
15        r=0;
16        while(s){
17            r=r*10+(s%10);
18            s=s/10;
19        }
20        if(ts==r)
21            break;
22        n=ts;
23
24    }
25    printf("%lld",ts);
26    return 0;
27
28 }
```

	Input	Expected	Got	
✓	32	55	55	✓
✓	789	66066	66066	✓

Passed all tests! ✓

Question 3

Correct

Marked out of 7.00

Flag question

Objective :

A number is considered lucky if it contains either 3 or 4 or 3 and 4 both in it. Write a program to print the nth lucky number. Example, 1st lucky number is 3, and 2nd lucky number is 4 and 3rd lucky number is 33 and 4th lucky number is 34 and so on. Note that 13, 40 etc., are not lucky as they have other numbers in it.

The program should accept a number 'n' as input and display the nth lucky number as output.

Sample Input 1:

3

Sample Output 1:

33

Explanation:

Here the lucky numbers are 3, 4, 33, 34., and the 3rd lucky number is 33.

Sample Input 2:

34

Sample Output 2:

33344

Answer: (penalty regime: 0%)

```

1  #include<stdio.h>
2  int main(){
3      long int i,j;
4      int r,n,c=0,f;
5      scanf("%d",&n);
6      for(i=1;c<=n;i++){
7          f=0;
8          j=i;
9          while(j>0){
10             r=j%10;
11             if(r==3||r==4)
12                 j=j/10;
13             else{
14                 f=1;
15                 break;
16             }
17         }
18         if(f==0){
19             c++;
20             if(c==n)
21                 break;
22         }
23     }
24     printf("%ld",i);
25 }

```

	Input	Expected	Got	
✓	34	33344	33344	✓

Passed all tests! ✓